

I run the following factor regressions:

$$\begin{aligned} r_{it} &= \alpha + \beta_{REIT}REIT_t + \beta_b r_{bt} + \varepsilon_{it}, \\ r_{it} &= \alpha + \beta_b r_{bt} + \beta_s r_{st} + \varepsilon_{it}, \\ r_{it} &= \alpha + \beta_{REIT}REIT_t + \beta_b r_{bt} + \beta_s r_{st} + \varepsilon_{it}. \end{aligned} \tag{10.13}$$

where  $REIT_t$  is the return to the NAREIT portfolio consisting of traded REITs,  $r_{bt}$  is the bond return, and  $r_{st}$  is the stock return, which have factor loadings of  $\beta_{REIT}$ ,  $\beta_b$ , and  $\beta_s$ , respectively. We require that, in all cases, the factor loadings add up to one so that they can be interpreted as a factor portfolio benchmark.

The estimated coefficients are:

|               | Coefficient | T-stat | Coefficient | T-stat | Coefficient | T-stat |
|---------------|-------------|--------|-------------|--------|-------------|--------|
| Alpha         | -0.51%      | -1.02  | -0.43%      | -0.90  | -1.50%      | -1.05  |
| REIT Loading  | 0.30        | 5.92   |             |        | 0.12        | 1.81   |
| Bond Loading  | 0.70        | 14.0   | 0.65        | 12.7   | 0.26        | 3.75   |
| Stock Loading |             |        | 0.35        | 6.95   | 0.61        | 11.6   |

For all of these factor benchmarks, direct real estate does not offer significant returns in excess of a factor benchmark. In fact, the point estimates are negative

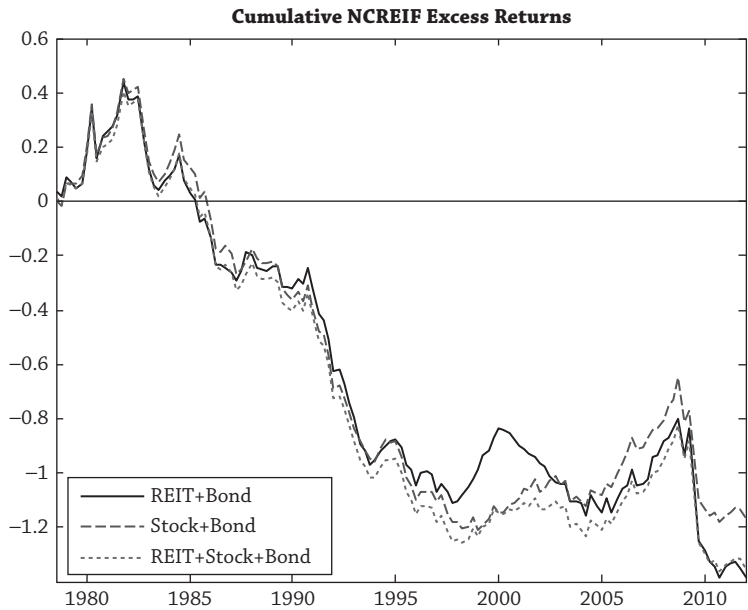


Figure 10.4